

Sadeeq Adeyemi

Computer Vision & Perception Engineer

Multi-Object Tracking · Sensor Fusion · State Estimation · Deep Learning for Robotics & Autonomous Systems

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PROFILE

Perception engineer working on one problem: **keeping track of what a robot cannot see**. I build occlusion-aware multi-object trackers that fuse camera, lidar and radar, and back every claim with multi-seed ablations.

RESEARCH

Occlusion-Aware Multi-Object Tracking for Lunar Surface Operations

2026 · independent

Python · PyTorch3D · MATLAB · JavaScript / Three.js — github.com/sadeeqgandalf/lunar-occlusion-tracking

- **Method.** 3-D visibility model (line of sight against boulders, polar cast shadows) gives each track an expected detection probability, used in an EKF + Hungarian + Bernoulli-existence tracker with a negative-information update, $p(x | \text{miss}) \propto p(x) (1 - P_D(x))$.
- **Results** (20 seeds $\times$ 300 s, 95% intervals). Identities kept through 1–5 s occlusions: **0% naive** $\rightarrow$ **46%** $\rightarrow$ **89% with appearance re-ID**. ID switches per run: 75.6 $\rightarrow$ **6.6** fusing camera, lidar and a second camera.
- **Finding.** Adding sensors to a tracker with no visibility model raises ID switches 3.4 $\times$ (75.6 $\rightarrow$ 254); a second viewpoint helps more than a better sensor on the same mast.
- **PyTorch3D.** Built from source on Apple Silicon; detections taken from rendered pixels (silhouette visibility, depth-map range, colour signature), with a windowed render $\sim 20\times$ faster and exact. Re-ID lifts same-ID after hides over 5 s from 8% to 50%; particle PHD filter maps where hidden people can be.
- **Verification.** Python port matches the JavaScript engine exactly; checked against MathWorks GNN, JPDA and TOMHT trackers.

Closed-Loop Autonomous Driving Stack

2026

- EKF localization, SAM 3 open-vocabulary perception lifted to 3-D on lidar, radar, multi-object tracking, Frenet planning, LQR/PID control, safety monitor; never sees simulator ground truth. **100/100** Monte Carlo runs pass; without occlusion reasoning an occluded pedestrian is struck in **5 of 10**.

Further perception work

2025 – 2026

- **Open-vocabulary MOTs:** SAM 3 text-prompted segmentation and tracking on KITTI and BDD100K (TrackEval, HOTA).
- **Radar from raw signals (MATLAB):** FMCW IF samples $\rightarrow$ range-Doppler $\rightarrow$ CFAR $\rightarrow$ angle, checked against the CRLB.
- **Lidar on Argoverse 2:** DBSCAN clustering, PCA boxes, moving-object segmentation from SE(3) ego-motion.
- **Estimation and deep learning:** EKF / UKF / particle filters under faults; ViT and NeRF from scratch in PyTorch.

EXPERIENCE

Robotics Research Assistant, University of Maine Robotics Laboratory

2025 – present

- Build mobile robot platforms (sensors, wiring, ROS 2 nodes); EKF localization fusing IMU, wheel odometry and lidar; costmap-based planning and obstacle avoidance; point-cloud perception pipelines.

Machine Learning Engineer and Structural Steel Fabricator, NewTech Steel, Nigeria

2018 – 2023

- Deployed a computer vision model for real-time fatigue and defect detection in steel members; automated testing cut test time by 50%. Led structural steel fabrication and erection for factory and auditorium projects.

Materials Engineer, R.W. Gillespie, United States (CPT): materials testing to ASTM standards

2024

EDUCATION

MS, Data Science & Engineering, UMaine, 2025 – present
MSc, Project Management, Univ. of Portsmouth, UK, 2023

MS, Structural Engineering, UMaine, 2023 – 2025
BEng, Civil Engineering, FUT Akure, Nigeria, 2019

SKILLS

Tracking & estimation	EKF / UKF, particle and PHD filters, GNN, JPDA, MHT; CLEAR-MOT, IDF1, HOTA, GOSPA
Vision & deep learning	PyTorch, PyTorch3D, SAM 3, YOLOv8, Vision Transformers, NeRF, re-ID, Open3D, OpenCV
Robotics & tools	ROS 2, Python, C++, MATLAB, JavaScript, Git, Docker, AWS (certified: ML Specialty, SA Associate)